

Projection to 2045: Environmental and Economic Performance of MSW Recovery Pathways

Six options for 21 Indonesian cities and regencies in 2045 against the existing condition of 2025: greenhouse gases, fossil energy, net cost, abatement cost and the preferred option, with a grid decarbonising linearly to net zero in 2060 and real escalation of electricity (2%/yr), labour-driven O&M (1%/yr) and landfill cost (2%/yr) — MSW pathway model v3.2

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Key findings: environmental performance

- By 2045 the grid emits 57% less per kWh. This is the only driver that changes emissions per tonne.
- **WtE loses most:** its GHG rises by 177 kg CO₂e/t (median), from 130 to 296, because each exported kWh displaces less coal-based power; its GHG avoided against the landfill falls from 441 to 238 kg CO₂e/t.
- **AD alone** also loses credit (+11 kg/t). **RDF improves** (-20 kg/t) because its electricity use becomes cleaner while the coal it displaces in kilns does not change; **RDF + AD** is almost unchanged (-1 kg/t).
- The **landfill** and **PHB** do not change: their emissions do not involve grid electricity.
- RDF + AD avoids more GHG against the landfill than WtE in 18 of 21 locations in 2045, against 7 of 21 in 2025. Climate benefits of energy recovery that rely on displacing grid power shrink as the grid decarbonises; benefits that rely on keeping organics out of landfills and displacing kiln coal persist.
- Fossil-energy savings follow the same pattern: WtE saves 1.7 GJ/t in 2045 against 4.2 GJ/t in 2025.

Key findings: economic performance

- Net cost of the **landfill** rises from 18.4 to 25.9 USD/t (median): landfill escalation outweighs the economies of scale of larger sites.
- **WtE becomes cheaper** (57.9 → 50.8 USD/t): electricity revenue grows by 49% and larger plants are cheaper per tonne. Locations reaching 1,000 t/day (Perpres 109/2025 tariff) rise from 6 to 8.
- **RDF, AD and RDF + AD** become dearer by about 5–8 USD/t, mainly through O&M and residue-landfill escalation.
- The **abatement cost** of RDF + AD against the landfill stays near 72 USD/t CO₂e (2025: 73); that of WtE stays near 98 because its lower cost is offset by its smaller climate benefit.
- **The preferred option hardly changes:** landfill up to 50 USD/t in both years; at 100 USD/t RDF + AD is most probable in 19 locations in 2045 (2025: 18). The decision rules of the main study remain valid under these drivers.

1 What is projected and how

The six options are evaluated per tonne of mixed MSW at the facility gate for the existing condition of 2025 and for 2045, for the same 21 locations, with the same model, parameters and composition. Three drivers change between the two years (Table 1); everything else is held constant so that the difference isolates their effect. Both years use the same Monte Carlo random numbers, so changes are computed draw by draw (paired).

$$EF_{grid}(t) = EF_{grid}(2025) \max[0, 1 - \delta(t - 2025)], \quad \delta = \frac{1}{35} \text{ yr}^{-1} \Rightarrow EF_{grid}(2045) = 0.429 EF_{grid}(2025) \quad (1)$$

$$x(t) = x(2025) (1 + e_x)^{t - 2025}, \quad Q(t) = Q(2025) (1 + g)^{t - 2025} \quad (2)$$

Driver	Rule	Factor 2025 to 2045	Affects
Grid emission factor	linear to zero in 2060 ($\delta = 1/35$ per year)	$\times 0.429$	electricity credits of WtE and AD, electricity use of RDF and front-end sorting, fossil CED of grid electricity
Electricity value	real escalation 2%/yr	$\times 1.486$	revenue of WtE and AD
Labour-driven O&M	real escalation 1%/yr	$\times 1.220$	O&M of WtE, RDF and AD plants
Landfill cost	real escalation 2%/yr (third rate, see note)	$\times 1.486$	sanitary landfill and open-dump cost, including residue landfills of every option
Tonnage	each location's growth rate; composition constant	median $\times 1.41$	plant scale (economies of scale) and scale gates

Table 1: Drivers of the projection (data/projection_2045_parameters.csv). CAPEX is constant in real terms. The third escalation rate of the 2/1/2% set is applied to landfill cost, an interpretation to be confirmed. Grid factors in 2045: Jamali 0.877 → 0.376, Sumatera 0.832 → 0.357, Mahakam 1.128 → 0.483 kg CO₂/kWh.

Environmental indicators. Only the grid driver changes greenhouse gases and fossil energy per tonne: escalation changes prices, not physical flows, and per-tonne emissions do not depend on plant size in this model. Landfill

methane, fossil carbon of plastics and coal displaced in cement kilns are unchanged; a cleaner grid makes electricity exported by WtE and AD displace less CO₂, and makes electricity consumed by RDF lines cleaner.

Economic indicators. All three drivers change costs: escalation raises O&M, landfill cost and electricity revenue; tonnage growth lowers the unit cost of capital through economies of scale (exponent $b = 0.7$) and of landfilling (elasticity e_{SL}); the grid affects cost only indirectly, through the ranking by carbon-inclusive cost.

Part A — Environmental performance

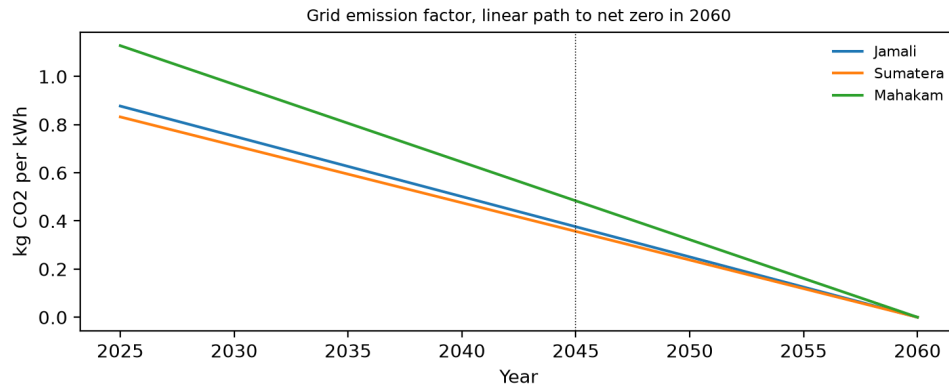


Figure 1: Grid emission factors of the three Indonesian systems on the assumed path (ESDM 2018 values in 2025).

2 Greenhouse gases per tonne, 2025 and 2045

Option	G 2025	G 2045	Change	Change %	Avoided vs OD 2025	Avoided vs OD 2045	Avoided vs SL 2025	Avoided vs SL 2045
SL landfill + flare	557 (424 to 793)	557 (424 to 793)	+0	+0%	425	425	-	-
S1 WtE	130 (-29 to 238)	296 (95 to 454)	+177	+143%	880	675	441	238
S2 RDF	319 (136 to 421)	299 (116 to 401)	-20	-7%	661	681	251	271
S3 AD	389 (192 to 696)	401 (210 to 700)	+11	+2%	582	573	161	150
S4 PHB	543 (414 to 773)	543 (414 to 773)	+0	+0%	439	439	14	14
S5 RDF + AD	153 (23 to 304)	150 (16 to 296)	-1	-1%	810	810	408	409

Table 2: GHG per tonne of MSW (kg CO₂e/t, GWP100), central values, market case: median (minimum to maximum) over 21 locations; avoided = baseline minus option (medians). Change and change % are medians of the per-location changes.

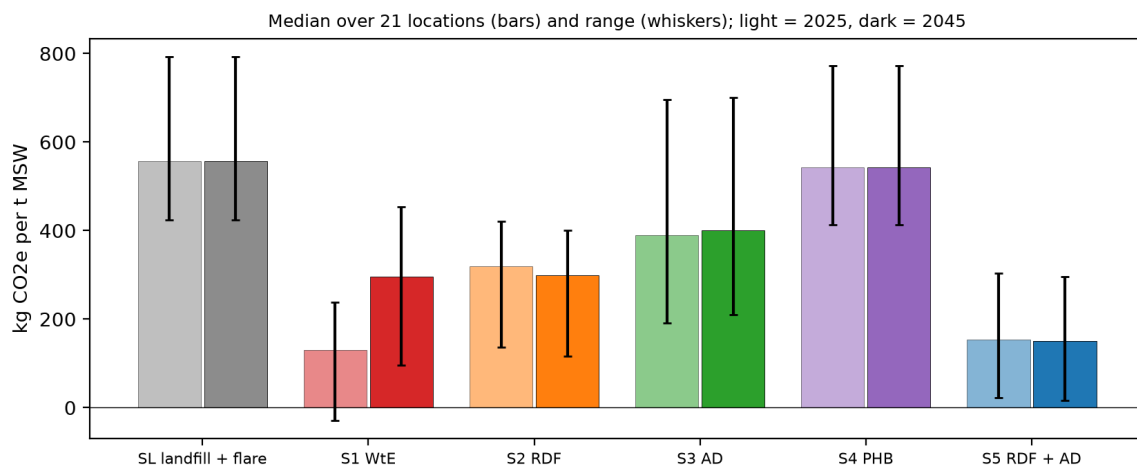


Figure 2: GHG per tonne in 2025 and 2045.

Option	Grid driver	Cost drivers	Tonnage	Total change
SL landfill + flare	+0.0	+0.0	+0.0	+0.0
S1 WtE	+177.5	+0.0	+0.0	+177.5
S2 RDF	-20.0	+0.0	+0.0	-20.0
S3 AD	+10.6	+0.0	+0.0	+10.6
S4 PHB	+0.0	+0.0	+0.0	+0.0
S5 RDF + AD	-1.4	+0.0	+0.0	-1.4

Table 3: Decomposition of the change in GHG (kg CO₂e/t, median over locations), each driver applied alone. Escalation and tonnage do not change emissions per tonne.

Option	Change p5	Change median	Change p95	Locations where GHG rises (median change > 0)
SL landfill + flare	-0	+0	+0	0
S1 WtE	+110	+167	+253	21
S2 RDF	-29	-20	-12	0
S3 AD	-1	+8	+21	21

Option	Change p5	Change median	Change p95	Locations where GHG rises (median change > 0)
S4 PHB	-0	+0	+0	0
S5 RDF + AD	-16	-4	+10	4

Table 4: Uncertainty of the change (paired Monte Carlo, 4,000 draws per location): 5th, 50th and 95th percentile of $G(2045) - G(2025)$, median over locations, kg CO₂e/t. The spread comes from the grid multiplier, efficiencies and composition.

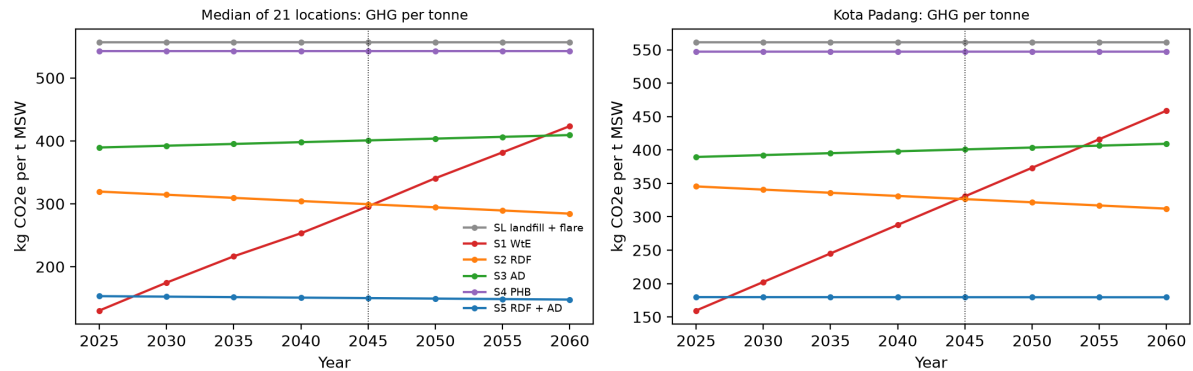


Figure 3: GHG per tonne from 2025 to 2060 on the assumed grid path (central values, all drivers).

3 Fossil cumulative energy demand

Option	CED 2025 (GJ/t)	CED 2045 (GJ/t)	Change (GJ/t)
SL landfill + flare	0.08	0.08	+0.00
S1 WtE	-4.20	-1.70	+2.50
S2 RDF	-3.80	-4.08	-0.28
S3 AD	-0.17	-0.02	+0.15
S4 PHB	-0.57	-0.57	+0.00
S5 RDF + AD	-4.18	-4.23	-0.04

Table 5: Fossil CED per tonne (negative = saving), medians over 21 locations. The fossil primary energy of grid electricity is scaled with the same grid path, so savings from exported electricity shrink.

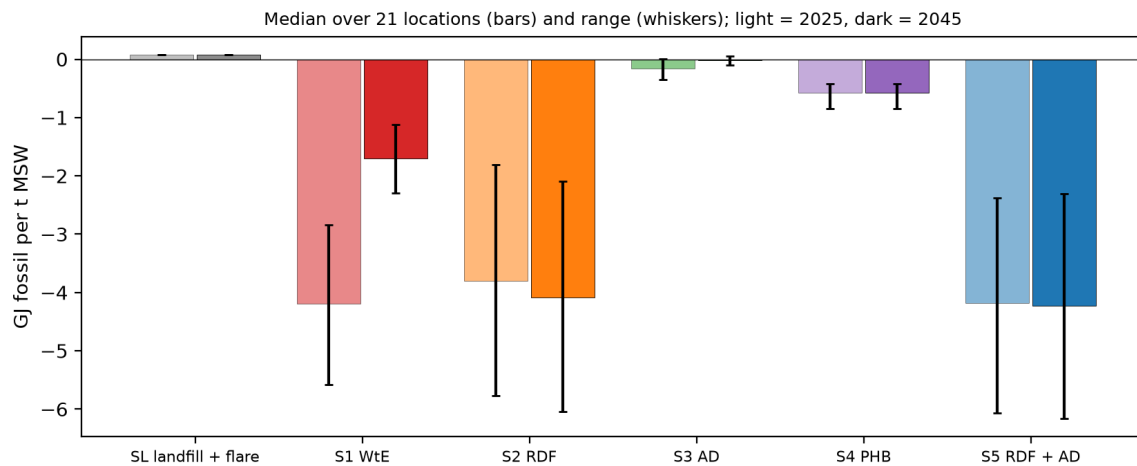


Figure 4: Fossil CED per tonne in 2025 and 2045.

4 Per-location environmental results

Location	S1 2025 / 2045	S2 2025 / 2045	S3 2025 / 2045	S5 2025 / 2045	Avoided vs SL: S5 - S1, 2045
Kota Padang	160 / 331	345 / 326	389 / 401	180 / 180	+151
Kab. Bogor	143 / 290	319 / 299	221 / 238	110 / 115	+175
Kab. Serang	238 / 435	345 / 324	261 / 275	159 / 161	+273
Kota Denpasar	-5 / 163	403 / 383	696 / 700	304 / 296	-133
Kab. Cianjur	92 / 241	345 / 325	377 / 389	176 / 176	+65
Kab. Indramayu	86 / 230	343 / 323	385 / 397	177 / 176	+54
Kota Semarang	130 / 308	296 / 276	384 / 393	153 / 150	+158
Kab. Brebes	237 / 454	254 / 234	335 / 340	141 / 134	+320
Kab. Nganjuk	56 / 207	414 / 394	505 / 515	258 / 256	-49

Location	S1 2025 / 2045	S2 2025 / 2045	S3 2025 / 2045	S5 2025 / 2045	Avoided vs SL: S5 – S1, 2045
Kab. Pati	171 / 408	136 / 116	341 / 346	23 / 16	+392
Kab. Magelang	102 / 285	267 / 247	398 / 407	124 / 121	+164
Kab. Pandeglang	116 / 284	289 / 269	192 / 210	68 / 74	+210
Kab. Pemalang	178 / 376	256 / 236	245 / 259	71 / 73	+303
Kab. Temanggung	135 / 300	307 / 287	372 / 378	182 / 177	+123
Kab. Pekalongan	155 / 364	292 / 272	446 / 455	151 / 148	+216
Kab. Sragen	111 / 296	296 / 276	412 / 422	148 / 145	+151
Kab. Kutai Kartanegara	-10 / 222	361 / 335	544 / 554	222 / 217	+5
Kab. Kendal	-29 / 95	421 / 401	481 / 498	211 / 216	-120
Kab. Wonogiri	13 / 163	342 / 322	440 / 456	141 / 145	+18
Kota Pekalongan	151 / 347	334 / 314	429 / 440	175 / 174	+173
Kota Magelang	142 / 346	272 / 252	457 / 464	145 / 140	+206

Table 6: GHG per tonne (kg CO₂e/t) for the grid-dependent options in each location, and the difference in GHG avoided against the landfill between RDF + AD and WtE in 2045 (positive = RDF + AD avoids more).

5 Environmental interpretation and limits

- Energy recovery evaluated with today's grid factor overstates its long-term climate benefit. A WtE plant commissioned around 2030 operates through the decarbonisation period, so its lifetime-average credit is closer to the 2045 value than to the 2025 value.
- Options whose benefit comes from avoiding landfill methane and displacing kiln coal (RDF, RDF + AD) keep their benefit; this is a robust argument for diversion of organics and RDF where a kiln is in reach.
- The landfill result assumes the same gas collection in both years; better collection or methane regulation would lower it, and kiln decarbonisation (less coal) would reduce the RDF credit. Neither is part of this projection.
- Composition is held at today's proxy; a richer, more plastic-intensive waste in 2045 would raise WtE emissions further.
- Screening-level results; all drivers are user-specified scenario assumptions, not forecasts.

Part B — Economic performance

6 Net cost per tonne, 2025 and 2045

Option	C 2025	C 2045	Change	Change %	Escalation	Tonnage scale	Interaction
SL landfill + flare	18.4 (11.8 to 40.0)	25.9 (15.1 to 53.7)	+6.6	+34%	+9.0	-1.8	-0.9
S1 WtE	57.9 (42.4 to 120.3)	50.8 (34.8 to 107.1)	-8.1	-13%	-2.3	-5.4	-0.3
S2 RDF	42.3 (31.3 to 70.2)	49.8 (36.9 to 81.2)	+7.8	+18%	+10.6	-2.4	-0.6
S3 AD	36.6 (28.1 to 65.8)	43.0 (30.7 to 76.8)	+5.6	+15%	+8.8	-2.5	-0.7
S4 PHB	38.1 (22.6 to 75.1)	43.2 (24.2 to 86.5)	+5.0	+13%	+9.0	-3.4	-0.9
S5 RDF + AD	47.1 (37.3 to 78.2)	52.7 (41.0 to 85.9)	+5.5	+12%	+8.7	-2.7	-0.5

Table 7: Net levelised cost per tonne (USD/t, real 2025 USD), central values, market case: median (range) over 21 locations, and the median change attributed to each driver applied alone.

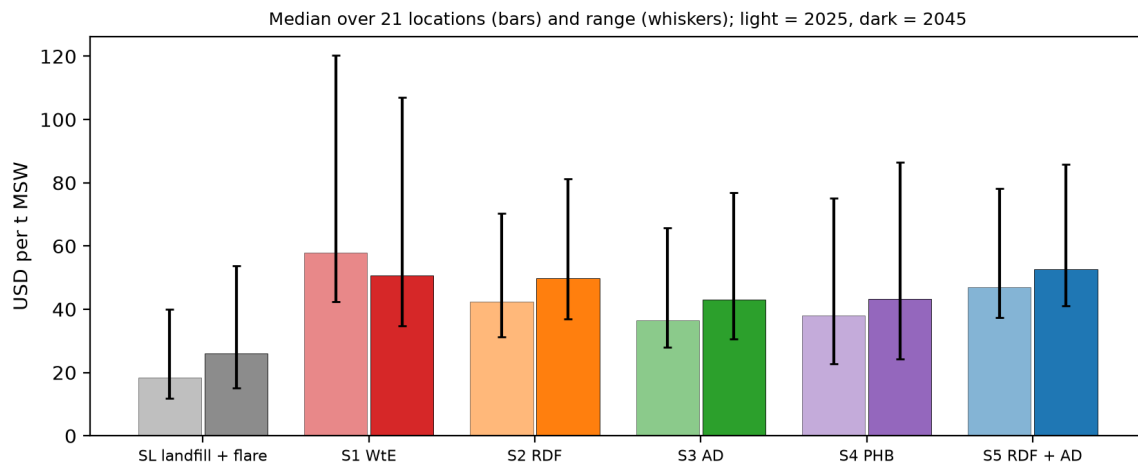


Figure 5: Net cost per tonne in 2025 and 2045.

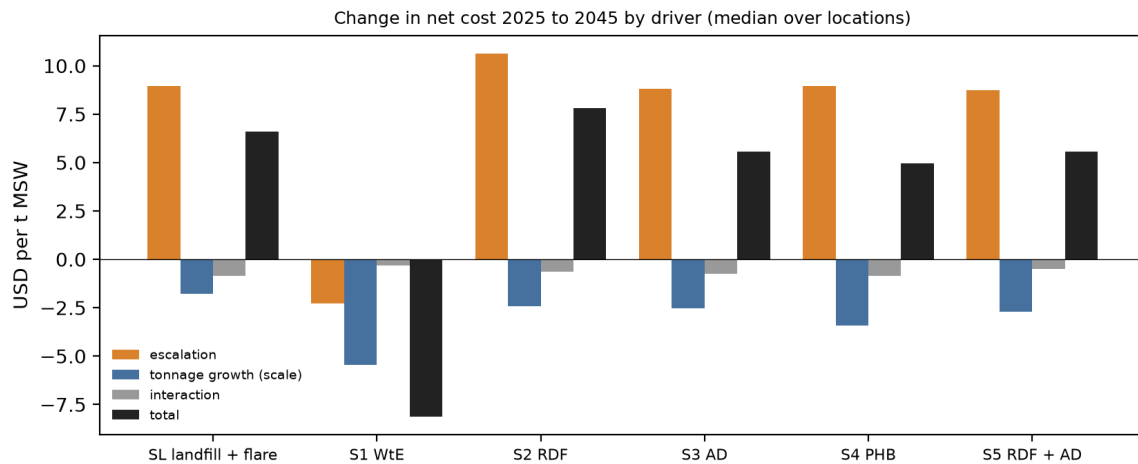


Figure 6: Decomposition of the change in net cost.

Option	Change p5	Change median	Change p95
SL landfill + flare	+4.0	+6.4	+8.9
S1 WtE	-16.3	-9.0	-3.8
S2 RDF	+5.7	+7.8	+9.9
S3 AD	+3.5	+5.7	+8.0
S4 PHB	+1.8	+4.6	+7.1
S5 RDF + AD	+3.8	+5.7	+7.6

Table 8: Uncertainty of the change in net cost (paired Monte Carlo, USD/t, median over locations).

7 Abatement cost and carbon-inclusive cost

Option	MAC vs SL 2025	MAC vs SL 2045	CIC at 50: 2025	CIC at 50: 2045	CIC at 100: 2025	CIC at 100: 2045
SL landfill + flare	-	-	45.8	53.0	72.7	78.9

Option	MAC vs SL 2025	MAC vs SL 2045	CIC at 50: 2025	CIC at 50: 2045	CIC at 100: 2025	CIC at 100: 2045
S1 WtE	98	98	64.4	64.1	70.1	80.2
S2 RDF	98	93	56.0	64.5	72.3	78.3
S3 AD	112	117	56.0	63.9	75.6	83.4
S4 PHB	1449	1347	65.1	70.7	90.4	95.0
S5 RDF + AD	73	72	54.0	59.8	60.9	66.9

Table 9: Abatement cost against the landfill (USD/t CO₂e) and carbon-inclusive cost $C + pG/1000$ (USD/t) at 50 and 100 USD/t CO₂e, medians over 21 locations.

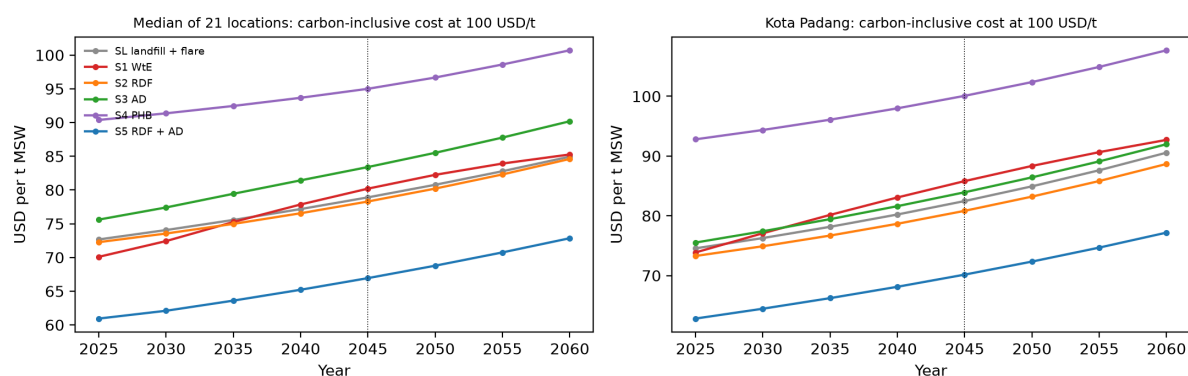


Figure 7: Carbon-inclusive cost at 100 USD/t CO₂e from 2025 to 2060 (central values, all drivers).

8 Preferred option at fixed carbon values

Carbon value (USD/t CO ₂ e)	Most probable 2025 (locations)	Most probable 2045 (locations)	Central best 2025	Central best 2045
0	SL 21	SL 21	SL 21	SL 21
2	SL 21	SL 21	as at 0	as at 0
25	SL 21	SL 21	SL 21	SL 21
50	SL 21	SL 21	SL 21	SL 21
100	S1 1, S5 18, SL 2	S5 19, SL 2	S1 1, S5 20	S1 1, S5 20

Table 10: Preferred option, 21 locations, market case (Monte Carlo 4,000 draws per location and carbon value; central values for the last two columns).

Location	Q 2025 t/d	Q 2045 t/d	P(S5) at 100: 2025	P(S5) at 100: 2045	P(SL) at 100: 2025	P(SL) at 100: 2045	Central best at 100: 2025 / 2045
Kota Padang	656	747	0.65	0.68	0.21	0.20	S5 / S5
Kab. Bogor	2,891	4,775	0.50	0.49	0.21	0.22	S5 / S5
Kab. Serang	1,187	1,696	0.32	0.36	0.37	0.37	S5 / S5
Kota Denpasar	918	1,302	0.47	0.49	0.07	0.06	S5 / S5
Kab. Cianjur	1,114	1,557	0.59	0.61	0.20	0.20	S5 / S5
Kab. Indramayu	1,152	1,442	0.62	0.63	0.21	0.22	S5 / S5
Kota Semarang	1,190	1,677	0.54	0.58	0.20	0.20	S5 / S5
Kab. Brebes	1,033	1,456	0.60	0.64	0.29	0.27	S5 / S5
Kab. Nganjuk	744	1,048	0.60	0.61	0.20	0.19	S5 / S5
Kab. Pati	691	974	0.83	0.85	0.08	0.06	S5 / S5
Kab. Magelang	671	946	0.62	0.65	0.19	0.18	S5 / S5
Kab. Pandeglang	527	742	0.41	0.45	0.36	0.34	S5 / S5
Kab. Pemasang	484	681	0.62	0.67	0.31	0.27	S5 / S5
Kab. Temanggung	423	596	0.41	0.46	0.48	0.44	S5 / S5
Kab. Pekalongan	560	789	0.66	0.71	0.12	0.10	S5 / S5
Kab. Sragen	402	567	0.62	0.67	0.20	0.17	S5 / S5
Kab. Kutai Kartanegara	373	526	0.00	0.00	0.45	0.48	S1 / S1
Kab. Kendal	437	616	0.67	0.68	0.20	0.18	S5 / S5
Kab. Wonogiri	422	595	0.70	0.73	0.15	0.14	S5 / S5
Kota Pekalongan	164	231	0.59	0.66	0.27	0.21	S5 / S5
Kota Magelang	50	70	0.50	0.65	0.37	0.24	S5 / S5

Table 11: Per-location results at 100 USD/t CO₂e (standard error of each probability ≤ 0.008).

9 Economic interpretation and limits

- Two effects cancel for WtE: higher electricity revenue and larger plants make it cheaper, while the cleaner grid removes much of its climate credit. Its position relative to RDF + AD therefore changes little, and it remains dependent on the tariff and on scale.
- Rising landfill costs make every option that reduces landfilling relatively more attractive, but by 2045 the effect is a few USD per tonne, smaller than the gap that the carbon value has to close.
- The 1,000 t/day gate of Perpres 109/2025 is reached by more locations as waste grows; whether that waste is collected is a separate question (managed shares are far lower today).
- CAPEX is constant in real terms and technology learning is not modelled; learning would lower the cost of the newer options (AD, RDF + AD, PHB). The Perpres tariff is held constant in real terms.
- Escalation rates and the grid path are user-specified scenario assumptions; the third rate of the 2/1/2% set is applied to landfill cost (to be confirmed). The parameters are in `data/projection_2045_parameters.csv` and the analysis reruns with `python run_projection_2045.py`.

10 Joint reading: what changes for the decision

- Environmentally, the cleaner grid erodes the credit of options that export electricity (WtE, AD) and slightly improves options that consume it (RDF); landfill-based options do not change.
- Economically, escalation makes landfilling dearer and electricity more valuable, and growing tonnage brings economies of scale; WtE becomes cheaper while the other recovery options become slightly dearer.
- The two effects largely offset each other for WtE, whose abatement cost stays near 98 USD/t CO₂e, while RDF + AD keeps both its climate benefit and its abatement cost (about 72 USD/t CO₂e).
- The preferred option is stable: landfill up to 50 USD/t CO₂e in both years; at 100 USD/t RDF + AD is most probable in 19 of 21 locations in 2045 (2025: 18). The decision rules derived for 2025 therefore remain valid for plants that operate into the 2040s.
- For stakeholders, a WtE business case should be evaluated with a declining grid factor over the plant's life, not with today's value; the climate case for RDF + AD and for source-separated AD does not depend on the grid.